A Project Report on

MindMaths: A StepWise Solution Generator for Engineering Mathematics Problems Using OCR

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "MindMaths: A StepWise Solution Generator for Engineering Mathematics Problems Using OCR" is a bonafide work of **Kartik Parmar(22106101), Niraj Patel(22106050), Dhruv Wesavkar(22106016), Rohan Patil(22106044)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Abstract

Engineering Mathematics (EM1–EM4) is a fundamental subject in the University of Mumbai curriculum, yet many students face challenges due to complex problem-solving techniques, manual calculations, and difficulty in understanding concepts. This project proposes a Mathematics Learning and Doubt-Solving Platform aimed at simplifying learning and improving students' conceptual clarity.

The system enables users to upload mathematical problems through images, screenshots, or text input. These inputs are processed using Optical Character Recognition (OCR) to accurately extract mathematical expressions. The extracted data is then analysed using a symbolic computation engine that generates detailed, step-by-step solutions, allowing students to understand the logic and methodology behind each problem rather than simply obtaining answers.

Additionally, the platform includes topic-wise learning modules, practice exercises, and progress tracking features to support self-paced and guided learning. An interactive and user-friendly interface provides real-time feedback, making the system function as a personalized digital tutor for engineering students.

This project aligns with **SDG 4: Quality Education** by promoting accessible, technology-driven learning tools that enhance understanding and academic performance. It also supports **SDG 9: Industry Innovation and Infrastructure** by integrating artificial intelligence, machine learning, and symbolic computation to modernize the educational ecosystem.

Overall, the platform bridges the gap between theoretical knowledge and practical application, helping students develop strong problem-solving skills and confidence in engineering mathematics.

Keywords - Engineering Mathematics, Optical Character Recognition (OCR), Symbolic Computation, Machine Learning, Convolutional Neural Network (CNN), LaTeX Conversion.

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ABBREVAION

AI	*Artificial Intelligence*
OCR	*Optical Character Recognition*
EM	*Engineering Mathematics*
MU	*Mumbai University*
ITS	*Intelligent Tutoring Systems*
MLP	*Mathematical Language Processing*
NLP	*Natural Language Processing*
CNN	*Convolutional Neural Networks*
RNN	*Recurrent Neural Networks*
GNN	*Graph Neural Networks*
LSTM	*Long Short-Term Memory*
MER	*Mathematical Expression Recognition*
CAS	*Computer Algebra System*
LLMs	*Large Language Models*
GPT	*Generative Pre-Trained Transformer*
BKT	*Bayesian Knowledge Tracing*
LCA	*Latent Class Analysis*
VLM	*Vision Language Model*
IM2LATEX	*Image to LaTeX Dataset*
WBMI	*Web-Based Mathematics Instruction*
API	*Application Programming Interface*
GPU	*Graphics Processing Unit*
LaTeX	*Lamport TeX (Document Preparation System)*
JSON	*JavaScript Object Notation*
JPEG	*Joint Photographic Experts Group*
SQL	*Structured Query Language*
AST	*Abstract Syntax Tree*
SSD	*Solid State Drive*
HCI	*Human Computer Interaction*

Chapter 1

Introduction

1. Introduction

In the evolving landscape of digital education and computational intelligence, solving complex mathematical problems efficiently and accurately has become an essential requirement across engineering and scientific disciplines. Mathematics forms the foundation of problem-solving in fields such as electronics, mechanical design, control systems, and computer science. However, students and professionals often face significant challenges when dealing with lengthy analytical derivations, symbolic manipulations, and advanced mathematical transformations such as Laplace transforms, Fourier series, differential equations, and matrix computations. Manual solving not only demands time and precision but also carries the risk of human error. To address these issues, the project titled **"**MindMaths: A stepwise Solution generator for Engineering Mathematics Problems using OCR**"** aims to bridge the gap between manual problem-solving and computational automation by creating an intelligent, user-friendly platform capable of interpreting and solving mathematical expressions step by step. The system is designed to take both text-based and image-based inputs, process them using advanced Optical Character Recognition (OCR) and symbolic computation techniques, and generate precise, step-by-step mathematical solutions [1][2].

At the core of the project lies a combination of powerful Python-based libraries and frameworks that work cohesively to ensure accuracy and performance. PaddleOCR and EasyOCR are employed to extract mathematical content from images, enabling the recognition of both handwritten and printed equations. These OCR models are fine-tuned to detect mathematical symbols, operators, and notations that typical OCR engines struggle with. To enhance equation parsing and conversion into machine-readable formats, Pix2Text is integrated, which transforms recognized mathematical expressions into LaTeX or symbolic forms. Once the equation has been successfully extracted and structured, the SymPy library performs the symbolic computation solving algebraic expressions, evaluating integrals, differentiating functions, and executing transforms [3]. NumPy provides efficient numerical processing and array handling, while Pillow is used for image preprocessing tasks such as resizing, thresholding, and noise reduction to improve OCR accuracy [4]. Together, these technologies create a seamless pipeline that can interpret images containing equations, convert them into symbolic expressions, process them computationally, and present the solution in a clean, formatted, and educationally valuable way using MathJax rendering.

The project directly addresses multiple real-world problems prevalent in education, research, and industry. Firstly, it solves the challenge of interpreting complex handwritten or printed mathematical content. Students often study from handwritten notes, scanned assignments, or textbooks that require them to manually retype equations into calculators or symbolic solvers a process prone to syntax and typographical errors [2]. The proposed system eliminates this inefficiency by allowing users to simply upload an image, from which equations are automatically extracted and processed. Secondly, it tackles the lack of transparency and explainability in existing online solvers. Most available tools provide only final answers without showing the intermediate reasoning steps, which limits their educational value. In contrast, this platform emphasizes step-by-step symbolic solutions, enabling learners to understand each phase of the problem-solving process. Thirdly, it provides an effective means of bridging traditional handwritten learning with modern digital computation. In many Indian and international educational settings, students continue to rely on handwritten materials; integrating OCR with symbolic math brings these traditional resources into the digital domain, supporting hybrid learning environments.

The project also has significant implications for engineering professionals and researchers. Many technical problems in electrical, mechanical, and control engineering rely on symbolic analysis for verification and modelling. For instance, calculating transfer functions, solving system equations, and verifying analytical derivations are common tasks where this tool can

reduce manual workload and enhance accuracy. The platform's modular architecture allows future integration of domain-specific problem solvers and visual output generators, further extending its usefulness beyond academia. Moreover, the system promotes accessibility and inclusivity in learning. Students with disabilities or those who struggle with typing mathematical notations can rely on the image input or potential speech-to-text integration in later versions, ensuring that the tool benefits a broader user base.

From a technological and educational perspective, the Mathematical Problem-Solving Platform demonstrates how artificial intelligence, computer vision, and symbolic mathematics can be combined to create intelligent learning tools [3][5]. It not only automates the computation process but also enhances comprehension by guiding users through logical steps. Unlike conventional solvers that function as black boxes, this platform acts as a digital tutor, promoting active learning and reducing dependency on rote memorization [6]. Additionally, the modular design allows the inclusion of new mathematical modules, such as Beta and Gamma functions, Fourier transforms, and vector calculus, making it scalable for future expansions. The system can also serve as a foundation for research into automated theorem proving, equation recognition, and symbolic reasoning areas that form the backbone of AI-driven mathematics education.

Ultimately, this project addresses the growing demand for intelligent educational software that simplifies mathematical problem-solving while maintaining academic rigor and interpretability. By integrating OCR systems like PaddleOCR and EasyOCR with powerful symbolic computation through SymPy, supported by Pix2Text, NumPy, and Pillow, the system creates an end-to-end pipeline that transforms handwritten or printed equations into clear, explainable, and accurate solutions [7]. It helps reduce the barriers between traditional and digital education, supports error-free learning, and fosters conceptual understanding. Beyond academic applications, such a system can also be used in research, data analysis, and engineering modelling, showcasing the transformative potential of AI in applied mathematics. This project is thus an innovative step toward a smarter, more accessible, and computationally empowered future for mathematical learning and problem-solving.

Chapter 2

Literature Survey

2. Literature Survey

[1]. An Expression Tree Decoding Strategy for Mathematical Equation Generation

The paper titled "An Expression Tree Decoding Strategy for Mathematical Equation Generation" presents a method for automatically generating mathematical equations from natural language problems using structured expression trees. From this study, it is understood that traditional sequence-based models often fail to maintain mathematical correctness and operational hierarchy, whereas the proposed approach overcomes this limitation by representing equations in a hierarchical tree structure. This representation helps capture both syntactic and semantic relationships between mathematical components effectively. It is observed that the model combines neural network architectures, particularly encoder–decoder frameworks, with tree-based decoding strategies to generate mathematically valid equations. The experimental results indicate improved accuracy and better interpretability compared to conventional linear models. Furthermore, the study highlights that such structured approaches are highly beneficial for automated problem-solving and educational applications, where understanding the reasoning behind equations is important. Overall, the paper demonstrates that expression tree-based methods enhance the reliability, transparency, and performance of AI-driven mathematical systems.

[2]. Introduction to Mathematical Language Processing

The paper titled "Introduction to Mathematical Language Processing" discusses the concept of Mathematical Language Processing (MLP), which integrates natural language processing, symbolic computation, and machine learning to enable computers to understand and solve mathematical expressions. From this study, it is understood that unlike traditional NLP, which focuses on linguistic meaning, MLP emphasizes the structural, logical, and symbolic aspects of mathematical language. The paper highlights that MLP involves tasks such as equation recognition, parsing, semantic analysis, and symbolic reasoning. It is observed that MLP systems can process both textual and visual representations of mathematical expressions, including handwritten and printed formats, and convert them into machine-understandable forms for computation. Furthermore, the study indicates that this technology serves as a foundation for intelligent math solvers, tutoring systems, and educational tools that provide step-by-step explanations. Overall, the paper demonstrates that Mathematical Language Processing plays a significant role in bridging human mathematical reasoning with computational systems, thereby enhancing digital education, accessibility, and personalized learning in mathematics and engineering domains.

[3]. SymPy: Symbolic Computing in Python

The paper titled "SymPy: Symbolic Computing in Python" discusses the development and capabilities of SymPy, an open-source Python library for symbolic mathematics. From this study, it is understood that SymPy functions as a complete computer algebra system (CAS) implemented in pure Python, supporting operations such as differentiation, integration, equation solving, matrix manipulation, and expression simplification. The paper highlights that its modular design allows easy integration into larger scientific and educational applications without the need for external dependencies. It is observed that SymPy offers flexibility, transparency, and accessibility, making it suitable for both researchers and students. Additionally, the library supports LaTeX rendering, enabling clear and readable mathematical outputs for better understanding. The study also indicates that SymPy plays an important role in promoting interactive learning and computational thinking in educational systems. Overall, the paper demonstrates that SymPy provides a strong and efficient foundation for symbolic computation, making it a key component in intelligent math solvers and digital learning platforms.

[4]. A survey on handwritten mathematical expression recognition

The paper titled "A Survey on Handwritten Mathematical Expression Recognition" provides a comprehensive review of methods and advancements in recognizing handwritten mathematical content. From this study, it is understood that early approaches relied on rule-based parsing and symbol segmentation, whereas modern techniques utilize deep learning models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformer architectures. The paper highlights that mathematical expressions are inherently complex due to their two-dimensional structure, including superscripts, subscripts, and non-linear relationships between symbols, making recognition more challenging than standard text OCR. It is observed that encoder–decoder frameworks and attention mechanisms have significantly improved the accuracy of symbol recognition and structural parsing. The study also discusses benchmark datasets such as CROHME, which play an important role in evaluating and improving model performance. However, it concludes that despite considerable progress, achieving human-level accuracy in handwritten mathematical expression recognition remains a challenge. Overall, the paper demonstrates that continuous advancements in neural and multimodal models hold strong potential for improving mathematical recognition systems in educational and computational applications.

[5]. Effectiveness of Intelligent tutoring Systems: A Meta-Analytical Review

The paper titled "Effectiveness of Intelligent Tutoring Systems: A Meta-Analytical Review" presents a comprehensive analysis of 50 controlled studies evaluating intelligent tutoring systems. From this study, it is understood that intelligent tutoring systems significantly improve student performance, with a median effect size indicating an increase of 0.66 standard deviations in test scores, effectively raising performance from the 50th to the 75th percentile. The paper highlights that the effectiveness of these systems depends largely on how learning outcomes are measured, with better results observed in locally developed assessments compared to standardized tests, emphasizing the importance of alignment between instructional content and evaluation methods. It is also observed that studies with nonconventional control groups or flawed implementations reported smaller improvements, indicating that both experimental design and system quality influence outcomes. Furthermore, the review suggests that the success of intelligent tutoring systems is influenced by factors such as implementation accuracy and the nature of comparison groups. Overall, the paper demonstrates that intelligent tutoring systems can significantly enhance learning outcomes when properly designed and implemented, making them valuable tools in modern education.

[6]. Advancing Education through tutoring systems: A Systematic Literature Review

The paper titled "Advancing Education through Tutoring Systems: A Systematic Literature Review" examines the development and effectiveness of Intelligent Tutoring Systems (ITS) in modern education. From this study, it is understood that AI-driven tutoring platforms personalize learning experiences through adaptive feedback, learner modeling, and real-time performance assessment. The review highlights that these systems replicate the role of human tutors by identifying students' weaknesses and providing targeted instruction, thereby improving engagement and knowledge retention. It is also observed that technologies such as natural language processing, data mining, and machine learning enhance the interactivity and responsiveness of these systems. Furthermore, the study concludes that ITS can significantly improve learning outcomes, especially in STEM subjects, by enabling self-paced and personalized education. However, it also identifies challenges such as high development costs, data privacy issues, and the need for proper pedagogical alignment. Overall, the paper establishes that intelligent tutoring systems play a vital role in advancing adaptive and technology-driven education.

[7]. SOLVE IT (MATH SOLVER)

The paper "SOLVE IT (Math Solver)" presents an innovative approach to automating mathematical problem-solving by combining Optical Character Recognition (OCR), symbolic computation, and machine learning. The system enables users to input equations through text or images, which are then processed using OCR models such as PaddleOCR or EasyOCR to extract mathematical expressions. These are subsequently solved using symbolic libraries like SymPy, with results displayed in a step-by-step format for better conceptual understanding. The study highlights how the system can assist students and educators by reducing manual effort and improving accuracy in problem-solving. By integrating artificial intelligence and natural language processing techniques, SOLVE IT bridges the gap between theoretical learning and computational tools. The project demonstrates the growing importance of intelligent math solvers in academic contexts, especially in engineering and scientific education, where they serve as both learning aids and automated evaluators.

[8]. ConvMath: A Convolutional Sequence Network for Mathematical Expression Recognition.

The paper titled "ConvMath: A Convolutional Sequence Network for Mathematical Expression Recognition" presents a deep learning-based framework for converting visual

mathematical expressions into structured symbolic representations. From this study, it is understood that the proposed ConvMath model combines convolutional neural networks (CNNs) with sequence modeling techniques to extract spatial features from handwritten and printed mathematical notations. Unlike traditional rule-based OCR systems, the model uses convolutional encoders to capture complex spatial relationships among symbols, while a sequence decoder generates the corresponding LaTeX or symbolic output. It is observed that this approach enables end-to-end recognition of mathematical expressions with improved robustness against noise, handwriting variations, and layout distortions. Experimental results indicate that ConvMath achieves higher accuracy and better generalization compared to conventional OCR and attention-based models. Furthermore, the study highlights its importance in the field of Mathematical Expression Recognition (MER), providing an efficient and scalable solution for integrating visual perception with symbolic computation. Overall, the paper demonstrates that such advanced models form a strong foundation for intelligent equation-solving systems and educational AI applications.

[9]. Symbolic Computation Applications in Power Engineering Education

The paper titled "Symbolic Computation Applications in Power Engineering Education" explores the use of symbolic computation tools in enhancing engineering education. From this study, it is understood that tools such as SymPy, MATLAB's Symbolic Toolbox, and Maple help students model, analyze, and solve complex mathematical problems efficiently. The paper highlights that symbolic computation bridges the gap between theoretical derivations and numerical analysis by enabling visualization of mathematical relationships, automation of derivations, and verification of results. It is observed that these tools are particularly useful in power engineering applications such as circuit analysis, Laplace transforms, control systems, and transient stability problems. Furthermore, the study indicates that incorporating symbolic computation in education improves student engagement, accuracy, and conceptual understanding. Overall, the paper demonstrates that symbolic computing plays a vital role in developing analytical and computational skills, and it strongly advocates for its integration into engineering curricula to prepare students for modern, technology-driven industry practices.

[10].Mask & Match: Learning to Recognize Handwritten Math with Self Supervised Attention

The paper titled “Mask & Match: Learning to Recognize Handwritten Math with Self-Supervised Attention” introduces a self-supervised learning approach for recognizing handwritten mathematical expressions. From this study, it is understood that the proposed method reduces dependence on large labeled datasets by using a masking-and-matching strategy, where parts of handwritten equations are masked and the model learns to reconstruct or match the missing components. This approach enables the system to understand contextual relationships between symbols and their spatial arrangements. The paper highlights that the integration of attention mechanisms helps the model focus on important symbol relationships and the two-dimensional structure of mathematical notation. It is observed that the model achieves performance comparable to fully supervised methods while requiring significantly less labeled data. Furthermore, the study emphasizes that self-supervised and attention-based architectures improve generalization, robustness, and data efficiency in mathematical expression recognition. Overall, the paper demonstrates that such approaches are highly effective for developing advanced handwritten math recognition systems and contribute significantly to intelligent math-solving and educational AI applications.

[11].The effectiveness of web-based Mathematics instruction (WBMI) on K-16 students’ mathematics learning: a meta-analytic research

The paper titled “The Effectiveness of Web-Based Mathematics Instruction (WBMI) on K–16 Students’ Mathematics Learning: A Meta-Analytic Research” examines the impact of web-based instructional methods on students’ mathematics learning outcomes. From this study, it is understood that web-based mathematics instruction significantly improves students’ academic performance, problem-solving abilities, and motivation compared to traditional teaching approaches. The research highlights that interactive and technology-driven learning environments, which include multimedia content, adaptive feedback, and real-time assessment, enhance students’ conceptual understanding and engagement. It is also observed that factors such as instructional design quality, learner autonomy, and teacher involvement play a crucial role in determining the effectiveness of WBMI. Furthermore, the study emphasizes that well-designed web-based platforms provide flexibility, accessibility, and opportunities for self-paced learning. Overall, the paper demonstrates that web-based mathematics instruction serves as an effective complement to conventional teaching methods and supports the advancement of modern, technology-enabled education systems.

[12].A Survey of Mathematical Reasoning in the Era of Multimodal Large Language Models

The paper titled "A Survey of Mathematical Reasoning in the Era of Multimodal Large Language Models" provides an overview of advancements in artificial intelligence for mathematical understanding and reasoning. From this study, it is understood that multimodal large language models integrate text, image, and symbolic data to enhance mathematical problem-solving capabilities. The paper discusses model architectures such as GPT, PaLM, and Gemini, which utilize deep neural networks to process equations, diagrams, and textual inputs simultaneously. It highlights a shift from traditional text-based reasoning to multimodal approaches that combine visual perception with symbolic logic, enabling systems to interpret handwritten equations, graphs, and geometric figures effectively. It is also observed that these models improve the ability of AI systems to understand complex mathematical concepts across multiple input formats. However, the study identifies challenges such as maintaining symbolic accuracy, ensuring generalization, and improving interpretability of results. Overall, the paper demonstrates that multimodal AI models play a significant role in advancing intelligent mathematical reasoning and represent a key direction for future research in AI-driven education systems.

Table 1: Summary of Existing Systems

Year	Title	Algorithm	Dataset	Gap
2025	SOLVE IT (Math Solver)	For OCR– PyTesseract with a custom dataset AI Reasoning- DeepSeek 7B LLM for stepwise solutions.	Custom Dataset	Handles limited handwriting/ formatting; lacks performance data, domain coverage clarity, scalability details
2025	Advancing Education through tutoring systems: A	Bayesian Knowledge Tracing (BKT) for modelling student knowledge and	No new dataset. It is a review based on 86 selected studies	Ethical concerns, scalability issues, limited cognitive adaptability in RTS,

	Systematic Literature Review	adapting instructions in ITS. Large Language Model (LLMs) for conversational and adaptive feedback. Latent Class Analysis (LCA) to categorize tutors into 3 classes- Computer-based ITS, Robot-based RTS and multimodal.	via PRISMA framework	need for hybrid systems
2025	A Survey of Mathematical Reasoning in the Era of Multimodal Large Language Models	Survey of recent multimodal LLM methods — reasoning, chain-of-thought, tool-use integration	Public benchmarks: GSM8K, MATH, MathVista, MathGLM, etc.	LLMs still lack precision in symbolic manipulation; hybrid systems combining AI and symbolic solvers are underexplored.
2025	Mask & Match: Learning to Recognize Handwritten Math with Self-Supervised Attention	Self-supervised attention-based Transformer model with contrastive and masking pre-training	CROHME dataset (standard handwritten math dataset)	Improves handwriting recognition but lacks integration with solvers or educational feedback systems.

2024	A survey on handwritten mathematical expression recognition	Review of traditional structural parsing → deep encoder–decoder and GNN-based models	CROHME datasets (2014, 2016, 2019 competitions)	Focuses on recognition accuracy but not on semantic understanding or actual problem-solving after recognition.
2023	Introduction to Mathematical Language Processing	Symbolic representation methods, semantic parsing, and NLP-based processing for math language	Mathematical problem corpora and language-based math datasets (not standard like CROHME, more text-oriented datasets)	Current systems lack deep semantic understanding of mathematical language, struggle with context-based interpretation, and need better integration between natural language and formal math structures.
2023	An Expression Tree Decoding Strategy for Mathematical Equation Generation.	Expression tree decoding with layer-wise parallel expression generation and bipartite matching	MathQA, Math23K, MAWPS (multi-step equation datasets)	Scalability, hyperparameter tuning, lacks OCR/image front-end, domain specialization.
2022	The effectiveness of web-based Mathematics instruction	Meta-analysis of multiple empirical studies comparing	63 empirical studies with 115 effect sizes across K–16 levels	Lacks AI-driven, step-by-step problem-solving and interactive feedback

	(WBMI) on K-16 students' mathematics learning: a meta-analytic rese arch	web-based vs. traditional teaching		mechanisms for personalized learning.
2020	ConvMath: A Convolution al Sequence Network for Mathematica l Expression Recognition.	Convolutional Sequence Network (CNN + RNN hybrid model)	Public handwritten mathematical expression datasets (e.g., CROHME dataset	Existing models struggle with complex notations, structural ambiguity, and recognition accuracy. The paper improves performance but still faces challenges with noisy/irregular handwriting.
2017	Symbolic Computation Applications in Power Engineering Education	Conceptual/descript ive approach using symbolic tools (Mathematica, Maple, Mathcad) for problem solving and visualization	Educational course examples; no specific dataset used	Does not address unstructured input handling or integration of symbolic solvers into web-based educational tools.
2017	SymPy: Symbolic Computing in Python	Rule-based symbolic algorithms used like (Risch,	No dataset is being used since it is a rule-based system and not data driven.	Limited scope (no OCR/handwriting support, incomplete solvers.)

		factoring, Gröbner basis etc.)		
2016	Effectiveness of Intelligent tutoring Systems: A Meta-Analytical Review	Meta-analytic synthesis of effect sizes (no new algorithm)	Aggregated results from previous studies (no new data collection)	Test inconsistency, influence of implementation quality, lack of design feature insights

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Although several online platforms and mobile applications are available for learning mathematics, they have significant limitations when applied to Engineering Mathematics (EM1–EM4) under Mumbai University:

- Generic Content – Most platforms provide general mathematics tutorials and examples, not specifically tailored to the Mumbai University syllabus.
- Lack of OCR for Math – Many tools cannot accurately recognize mathematical equations from handwritten or scanned questions, limiting their usefulness for students.
- Fragmented Learning – Learning materials, practice questions, and doubt-solving are often spread across multiple apps rather than being integrated into one platform.
- Absence of conceptual guidance – Most solvers only provide final answers without step by-step explanations or conceptual reasoning. Because of this, students end up memorizing solutions rather than understanding Mathematical principles.
- Limited scope of problem solving – Man existing Math solvers (like SOLVE IT and Sympy-based tools) are restricted to specific types of equations or predefined symbolic forms. They often fail to interpret complex, handwritten, or multiple-step problems commonly found in Engineering Mathematics.

- Inadequate Real-Time Assistance – Many current systems are offline or slow to process image-based queries, reducing their usability as on-the-go learning tools. There is often no interactive interface for real-time doubt resolution.

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

Engineering Mathematics (EM1–EM4) is a core subject in the Mumbai University curriculum, yet many students struggle with understanding concepts and solving problems due to the subject's abstract nature and lack of personalized guidance. Traditional resources such as textbooks and lectures provide limited support for instant doubt resolution, while existing online tools are often tailored for a more generic syllabus and fails to process handwritten or scanned problems accurately.

To address this issue, we propose a Mathematics Learning and Doubt-Solving Platform. The system allows students to upload questions via images or screenshots, which are processed using Optical Character Recognition (OCR) to extract mathematical expressions. A symbolic computation engine then generates step-by-step solutions, ensuring conceptual clarity. The platform also includes topic-wise learning modules and progress tracking, offering personalized and interactive support. This solution acts as a digital tutor, reducing academic challenges and improving student outcomes.

4.2 Objectives

- To enable instant doubt solving by allowing students to upload mathematical questions through images or screenshots and receive accurate, step-by-step solutions.
- To implement Optical Character Recognition (OCR) for extracting mathematical expressions from handwritten and printed inputs efficiently.
- To provide detailed, step-by-step solutions that enhance conceptual understanding instead of giving only final answers.
- To develop topic-wise learning modules with structured study material, solved examples, and practice exercises aligned with the EM1–EM4 syllabus.
- To design a user-friendly and accessible platform that can be easily used by students on web and mobile devices.

4.3 Scope

The proposed Mathematics Learning and Doubt Solving Platform is designed to support students of Mumbai University in mastering Engineering Mathematics I, II, III and IV (EM1 EM4).

- Allow students to upload mathematical problems through images or screenshots.
- Use OCR technology to extract mathematical expressions from printed or handwritten content.
- Provide step-by-step solutions using symbolic computation, ensuring conceptual clarity rather than just final answers.
- Offer topic-wise study modules aligned with the Mumbai University syllabus.
- Include interactive quizzes and practice tests for self-assessment.
- Be accessible via a user-friendly interface on web and mobile devices.

Chapter 5

Proposed System

5. Proposed System

5.1 Methodology

The proposed system follows a structured methodology to support effective learning and doubt resolution in Engineering Mathematics (EM1–EM4). The methodology is designed to guide students through a clear problem-solving process, emphasizing conceptual understanding rather than direct answer retrieval.

Initially, the user selects the Engineering Mathematics module and then selects the type of problem needs to be solved. The user can either type the problem or upload questions via screenshots. These inputs are then preprocessed to enhance its readability, which then help to draw out accurate interpretation of mathematical expressions [8]. The extracted mathematical content is then converted into its corresponding LaTeX format for further analysis.

The system then maps the problem to the predefined solution strategies aligned with standard university level methodologies. Solutions are generated in a step-by-step manner, reflecting classroom-style derivations commonly expected in examinations [2]. This approach ensures consistency, correctness, and pedagogical clarity across problems.

Once the solutions are generated, they are formatted into readable mathematical expressions and presented through a web-interface [1]. In addition to solution delivery, the platform organizes problems into topic-wise modules enabling guided practice and self-paced learning. Through this systematic workflow, the proposed methodology effectively addresses real-world academic challenges faced by engineering students in mastering complex mathematical concepts.

5.2 System Architecture

The proposed system follows a structured architecture to ensure modularity, clarity, and efficient processing of Engineering Mathematics problems. The system handles both textual and image based inputs and then provide structured step-by-step solutions.

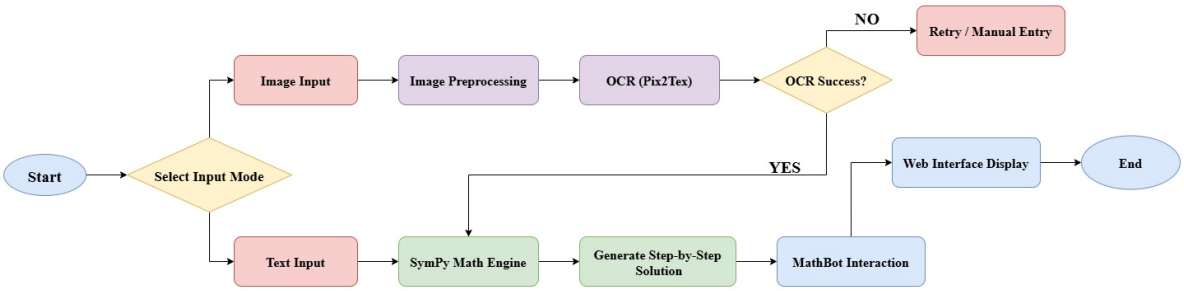

Figure 1: System Architecture

5.2.1 User Interface Layer

The user interface layer acts as the interaction module between the user and the system. It allows the users to either manually enter mathematical expressions or they can upload images containing handwritten or printed equations.

The system supports two modes of user input: text-based and image-based. When a user enters a mathematical expression in text form, it is first converted into a standardized LaTeX representation to ensure uniformity and accurate symbolic interpretation [4][8]. In case of image-based input, the image is first processed to extract the mathematical expression, which is later converted to its corresponding LaTeX form. The system converts all equations to its LaTeX form to establish a consistent intermediate representation for further processing.

$$f(x) = (x+1)(\sqrt{x}-1)(\frac{1}{x}+4) \longrightarrow$$

```
f(x)=(x+1)({\sqrt{x}}-1)({\frac{1}{x}}+4)
```

Input Equation LaTeX Equation

Figure 2: Image-To-LaTeX Conversion

5.2.2 Backend Layer

The Backend Layer is the central control unit of the proposed system. It receives input from either the user interface or the OCR module through a request handler, which is responsible for managing request routing, validation and error handling. This ensures that only well-formed and interpretable inputs proceed to subsequent stages to be processed.

An equation parser analyses the extracted text to determine the specific mathematical operation involved, such as integration, Laplace transform, or matrix computation. Based on the analysis, the extracted expressions are arranged into a standardized format, which is suitable for further computation.

5.2.3 Math Engine Layer

The Math Engine Layer performs the core computational tasks. Using symbolic computation techniques, the engine executes the required mathematical operations. Additionally, a step-by-step logic generator decomposes the solution into intermediate stages, enabling users to understand the complete problem-solving methodology rather than just the final result [5][9].

Finally, the Output Module formats the computed result and its corresponding step-by-step explanation into a readable mathematical form. The formatted output is sent back to the user interface, where it is displayed using mathematical rendering tools to ensure clarity, correctness, and ease of interpretation.

5.2.4 OCR Models

PaddleOCR

PaddleOCR is a general-purpose text recognition system that performs well on clear, printed text, commonly found in textbooks, PDFs, and lecture notes. For this project, it is mainly used to extract mathematical expressions from well formatted image inputs such as screenshots or scanned question papers. The structured approach used in this system, allows it to reliably detect text regions and convert them into readable equation strings. Although this approach is efficient and accurate for standard mathematical symbols and linear expressions, it fails to capture higher level mathematical notation such as nested fractions and superscripts. Despite its limitations, PaddleOCR is valuable, because of its dependable and fast text recognition for well-formatted Engineering Mathematics problems.

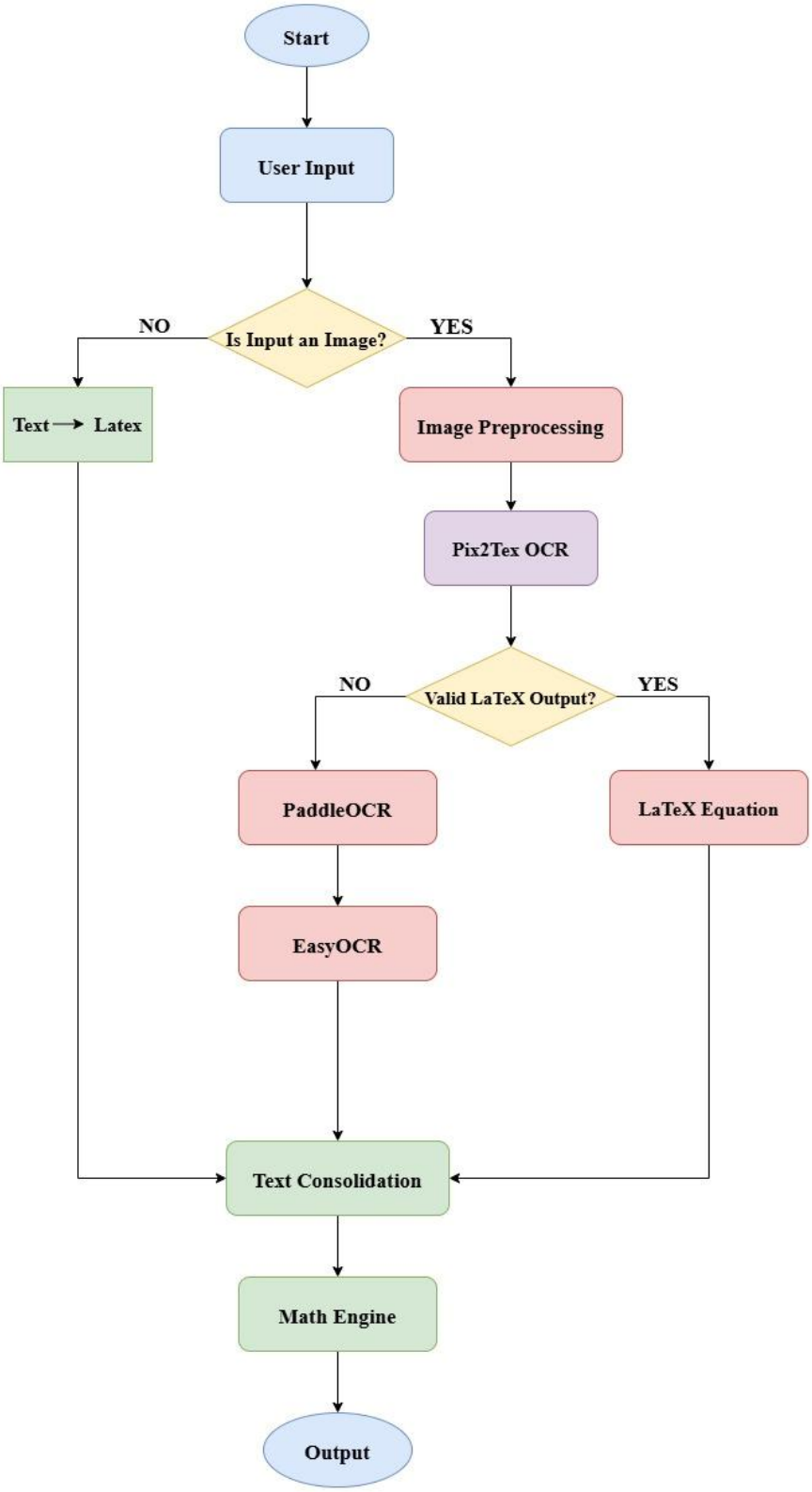

Figure 3: OCR Workflow

Pix2Tex

Pix2Tex is used to interpret mathematical expressions rather than general text, which makes it highly relevant towards our Engineering Mathematics learning platform. It converts plain text equations from images into its corresponding LaTeX format. This allows the system to

maintain a layout and meaning of complex expressions such as integrals, matrices, differential equations and transformations. For subjects like EM1 to EM4, where understanding symbolic structure is essential, Pix2Tex enables accurate parsing and step-by-step solution generation. Although it requires higher computational resources, its contribution significantly improves the correctness of the system, which in turn makes the system more reliable.

EasyOCR

EasyOCR is integrated into the system to handle inputs which are harder to interpret, particularly handwritten equations or images captured under poor lighting conditions. Since many students rely on their smartphones for clicking pictures of their classroom problems or handwritten notes, EasyOCR helps improves the system's robustness by recognizing irregular and noisy text. Although it has limited Mathematical understanding and it may fail to realize the structural relationship between the symbols, it serves as an effective fallback model, when other OCR models fail. This makes EasyOCR valuable as students can submit their doubts without having to worry about the image quality.

5.2.5 Integration of OCR Models in Proposed System

The proposed system integrates advanced OCR and deep learning models such as PaddleOCR, EasyOCR, and Pix2Text to accurately interpret and solve mathematical expressions from images and handwritten inputs. These models are combined in a structured pipeline to ensure both robustness and precision.

The workflow begins with input acquisition, where users upload images or handwritten mathematical expressions. The input then undergoes preprocessing steps such as noise removal, grayscale conversion, resizing, and binarization [8]. This improves image clarity and enhances OCR performance.

Next, PaddleOCR and EasyOCR are used for text detection and recognition. PaddleOCR primarily identifies text regions and extracts characters with high accuracy, even in complex or noisy images [10][9]. EasyOCR acts as a complementary model, improving recognition in cases where variations in handwriting or font styles may reduce accuracy. Using both models together increases reliability through cross-verification and fallback mechanisms.

However, general OCR models are not well-suited for capturing the two-dimensional structure of mathematical expressions. To address this limitation, the processed output is passed to Pix2Text, which specializes in mathematical OCR. Pix2Text converts the detected expressions

into structured LaTeX format while preserving spatial relationships such as fractions, superscripts, subscripts, and matrices [1].

The generated LaTeX code is then forwarded to the symbolic computation module, where it is parsed and solved using mathematical libraries. The system produces step-by-step solutions, which are displayed through a user-friendly interface [11].

Overall, the integration of PaddleOCR, EasyOCR, and Pix2Text creates a hybrid and efficient framework that combines general text recognition with specialized mathematical understanding, enabling accurate and scalable processing of engineering mathematics problems.

5.2.6 Chatbot

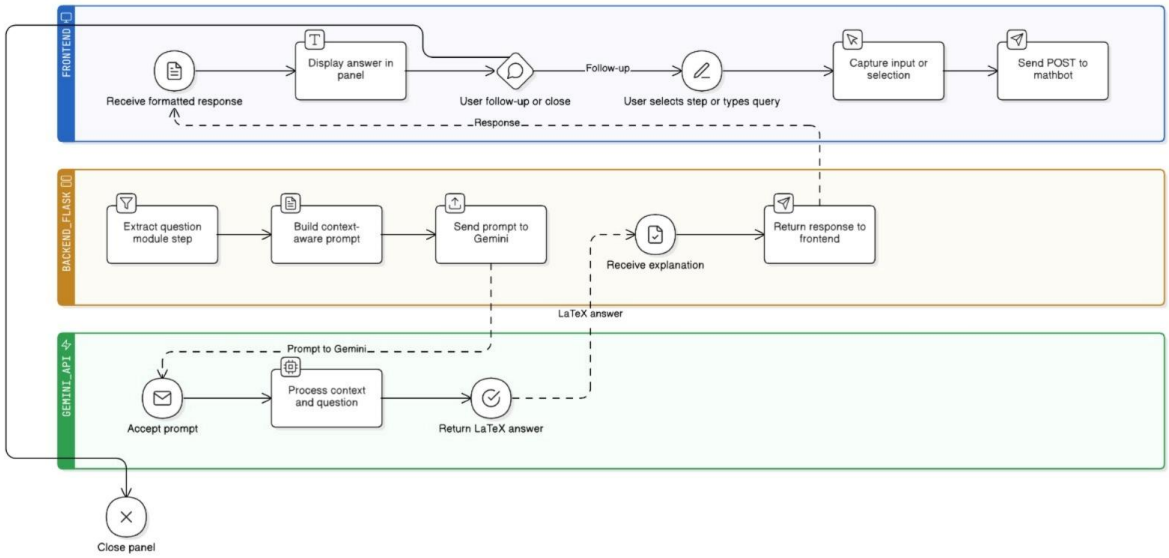

Figure 4: Chatbot Architecture

The chatbot in the proposed system is designed as a multi-layered architecture that integrates frontend interaction, backend processing, and AI-based response generation using the Gemini API. Its primary goal is to provide step-by-step explanations and interactive support for engineering mathematics problems.

The workflow begins at the frontend, where the user views the solved mathematical response and interacts with it. The user can either select a specific step from the solution or type a custom query (e.g., asking for clarification or explanation). This input is captured and sent to the backend via an API request.

In the backend layer, the system first extracts the relevant step or question from the user's input. It then constructs a context-aware prompt, which includes the original mathematical problem, the computed solution, and the specific step or query selected by the user. This ensures that the chatbot understands the context of the question rather than treating it as an isolated query.

The constructed prompt is then sent to the Gemini API. In the AI processing layer, Gemini processes the input by analyzing both the mathematical context and the user's query. It generates a detailed explanation or clarification, often returning the response in structured LaTeX format to preserve mathematical notation.

Once the response is generated, it is sent back to the backend, where it is post-processed and then forwarded to the frontend. The frontend displays the explanation in a formatted manner, allowing users to clearly understand each step.

The system also supports follow-up interactions, maintaining context across multiple queries. This enables users to ask deeper questions about specific steps without re-entering the entire problem.

Overall, the chatbot workflow ensures seamless interaction between user input, contextual understanding, and AI-generated explanations, making the learning process more intuitive, interactive, and effective.

5.3 Handling OCR Errors

The correctness of the mathematical expression extracted during the OCR stage determines the accuracy of symbolic computation in the suggested system [12][9]. In real-world situations, handwriting variations, image noise, or intricate mathematical layouts can occasionally cause OCR models to misinterpret symbols. The system incorporates a user validation step prior to the computation stage in order to address this problem. The interpreted equation appears on the user interface once the OCR module has extracted the equation and transformed it into the appropriate LaTeX representation. If any recognition errors are found at this point, the user has the option to manually edit or fix the equation. The equation is sent to the symbolic computation engine for step-by-step solution generation only upon user confirmation. This human-in-the-loop verification method guarantees greater reliability of the produced mathematical derivations and greatly lessens the impact of OCR recognition errors on the final

solution [1]. The system maintains accuracy and flexibility for various input formats, including handwritten notes, scanned question papers, and screenshots, by permitting user correction.

5.4 System Overview

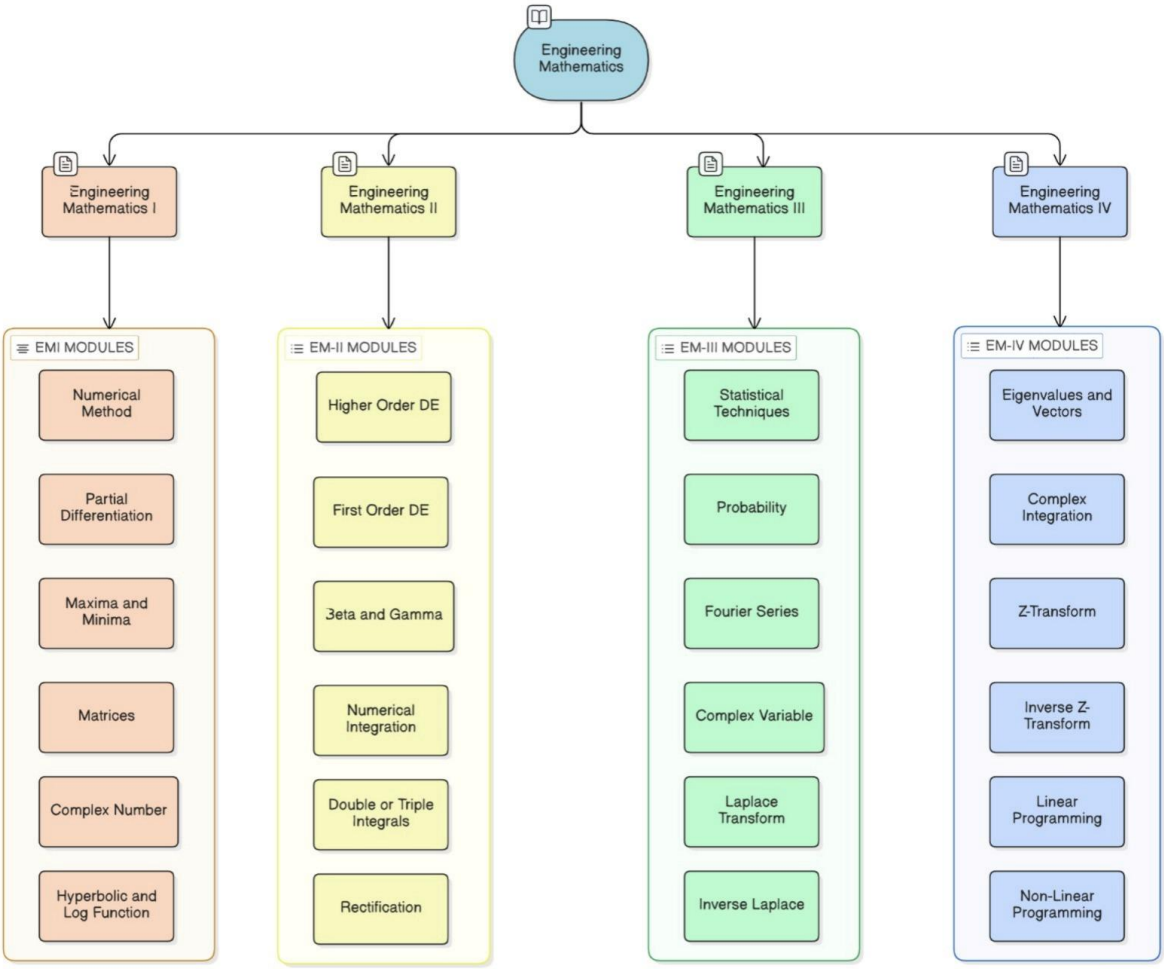

Figure 5: System Overview

This system is organized around the Mumbai University (MU) curriculum for the subjects Engineering Mathematics I, II, III, IV. Each of these subjects are having six different modules being taught in the undergraduate engineering program. These modules each cover a vast range of concepts spanning from differential equations, matrices, probability, Laplace transform etc. This helps the users to navigate concepts systematically. Also, in each module there is a dedicated formula sheet which is provided which provides the users a quick access to important formulas and key concepts for their revision and reference. This feature helps the users to quickly recall the fundamental concepts while solving problems, which further improves their problem-solving capabilities.

Chapter 6

Experimental Setup

6. Experimental Setup

Software Requirements:-

- Operating System: Windows 10/11, Linux or macOS
- Programming Languages:
  - Python (for data processing, ML model training, and backend)
  - HTML, CSS, JavaScript (for web-based user interface)
- Framework and Tools:
  - PaddleOCR, EasyOCR, Pix2Tex (for OCR and LaTeX expression recognition modules)
  - Jinja2, KaTeX (for UI templating and mathematical equation rendering)
  - SymPy (for symbolic mathematics computations and parsing)
  - PyTorch, Paddle (Deep learning backends for OCR models)
- Package Management:
  - pip, virtual env (dependency management and environment isolation)
- Version Control:
  - Git/GitHub (for code versioning and collaborative development)

- Browser Support:
    - Chrome, Edge, Firefox etc. (frontend display)

Hardware Requirements:-

- Processor (CPU): Intel Core i3(7th Gen or above) / AMD Ryzen 3
- GPU (optional): GTX 1050 or higher
- RAM: 8 GB
- Storage: 512 GB SSD

Operating Environment:-

- Web based application accessible via browsers on Windows, Linux, macOS.
- Students and educators can access via standard browsers.

Development Environment:-

- IDE Used: Visual Studio Code (VS Code).
- Image Processing Libraries: OpenCV, Pillow, and NumPy for preprocessing uploaded question images.
- OCR Frameworks: PaddleOCR, EasyOCR, and Pix2Tex (LaTeX OCR) for text and equation recognition.
- Mathematical Computation Engine: SymPy for symbolic processing and step-by-step solutions.
- Deep Learning Backends: PyTorch and PaddlePaddle for OCR model inference.
- Dependency Management: pip and virtualenv for isolated environments.
- Version Control: Git and GitHub for source code management.
- API Testing Utility: HTTPX for verifying backend endpoints.
- Environment Configuration: python-dotenv for handling environment variables securely.

Machine Learning & NLP Libraries:-

- PaddleOCR: A deep learning–based OCR tool used to detect and recognize text and mathematical symbols from images.
- EasyOCR: A lightweight OCR library that extracts handwritten or printed text using CNN and LSTM models.

- Pix2Text: A machine learning model that converts equation images into LaTeX or symbolic text format.
- SymPy: A Python library for symbolic mathematics that performs analytical solving and step-by-step computation.
- NumPy: A numerical computing library used for fast mathematical operations and matrix processing.
- Pillow (PIL): An image processing library used for resizing, cleaning, and enhancing images before OCR.
- Transformers: An NLP library that enables understanding and conversion of natural language mathematical queries into symbolic form.

Chapter 7

Results and Discussion

7. Results and Discussion

7.1 Dataset Used

Table 2: Dataset Characteristics and Statistics

Attributes	Details
Total Samples	≈ 103,558 image-formula pairs
Training Samples	≈ 90,000
Validation Samples	≈ 5,000
Test Samples	≈ 5,000
Image Resolution	160 x 320 pixels
Average Formula Length	≈ 30-40 LaTeX tokens
Vocabulary Size	≈ 500-700 unique LaTeX tokens

We have utilized the IM2LATEX-100K dataset, which is a large-scale, prebuilt benchmark dataset designed for image-to-LaTeX generation tasks. It contains approximately 100,000 paired samples, where each sample consists of a rendered mathematical formula image and its corresponding LaTeX source code. Unlike the traditional datasets, IM2LATEX-100k does not contain predefined numerical features, whereas these features are learned by the model automatically from receiving raw inputs:

- Image Features: Each sample contains rendered formula images. Pixel level data is extracted using Convolutional Neural Networks (CNNs) while model training, which helps the model capture the structural information of the Mathematical expressions such as the spatial alignment, symbols and Mathematical layout.
- Textual Features: The output is recorded as a sequence of LaTeX tokens, these tokens are then converted into their corresponding embeddings, where each token represents a learned feature from the high dimensional space.
- Feature Dimensionality: These are dependent on the model used and vary based on the architecture used in the CNN encoder and sequence decoder. Hence, the dataset itself does not have a fixed feature count.

7.2 Model Evaluation

Table 3: Performance Comparison of OCR Models

Metrics	PaddleOCR	EasyOCR	Pix2Tex
Character Recognition Accuracy (%)	92.4	78.1	96.8
Precision (%)	78.6	72.3	95.9
Recall (%)	91.2	76.4	95.1
F-1 Score	90.1	78.8	95.5
LaTeX Parsing Success Rate (%)	81.3	74.5	92.7

Because of its direct image-to-LaTeX decoding method, Pix2Tex performs better than the other general OCR models in every expression-level statistic, making it ideal for Engineering Mathematics tasks involving complex notations.

Although PaddleOCR has high character-level accuracy, it has trouble maintaining mathematical structure, especially when dealing with multi-line or nested formulas.

EasyOCR works well for simple text recognition, but it struggles to parse complex mathematical symbols and portray them accurately. Higher LaTeX parsing success rates directly correlate with improved symbolic computation reliability and step-by-step solution generation.

7.3 Experimental Results

Table 4: Performance Comparison of OCR and Image-to-LaTeX Models

Model	Accuracy	Precision	Recall	F-1 Score
TesseractOCR	67	65	58	61
InftyReader	87	88	82	85
Im2LaTeX	92	91	89	90
Mathpix	94	94	93	93
CNN + RNN	81	70	77	78
CNN + LSTM	80	78	79	78
PaddleOCR	88	88	84	86
EasyOCR	79	80	76	78
Pix2Tex	92	93	91	92

The main goal of the proposed system is to help students learn Engineering Mathematics by correctly interpreting the equations they send in and giving them step-by-step symbolic solutions. It also needs to keep the math structure, work with different types of input, and work with symbolic computation. Fractions, matrices, integrals, and summations are common two-dimensional layouts in engineering math problems. Pix2Tex changes pictures of equations into structured mathematical forms while keeping the spatial hierarchy and operator precedence.

This makes it better than regular OCR engines, which treat equations as linear text and often lose structural information.

There are high-accuracy systems like MathPix, but they need cloud access or proprietary APIs, which makes it hard to reproduce and use them for academic work offline. PaddleOCR and EasyOCR, on the other hand, are good open source options that work well on printed and semi-structured equations. They also make the system faster and more scalable. Students send in equations in a number of different ways, such as typed text, screenshots, and handwritten notes. No single OCR model works best in all situations, so combining general-purpose OCR engines with a math-aware recognition model makes it more robust against changes in handwriting, image quality, and formatting.

The symbolic computation engine (SymPy) used in the system works with the outputs of these models to make it possible to find solutions step by step. This method keeps things clear and helps the platform reach its educational goal of being a learning tool instead of a black-box solver.

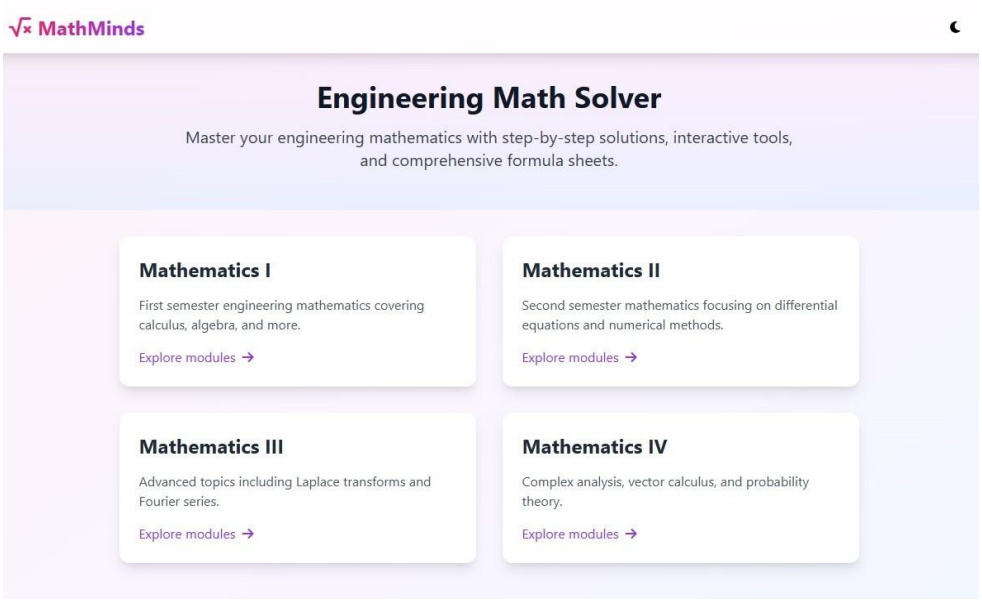

Figure 6: Web Interface Dashboard

After opening the MindMaths interface, a student can navigate through semester-wise engineering Mathematics module and can select their required subject.

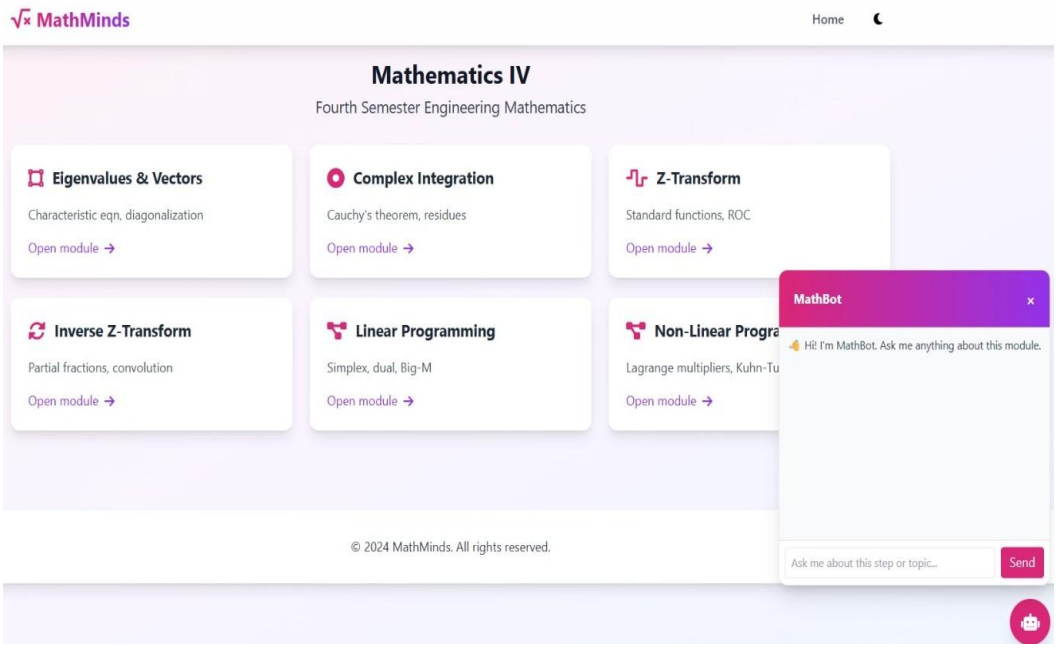

Figure 7: Web Interface for choosing preferred Module

After selecting Mathematics IV, the student is presented with all the major fourth-semester engineering Mathematics topics. From here on the student can choose a specific module such as Eigen vector and values, complex integration, Z-Transform, Inverse Z-transform, Linear programming and non-linear programming.

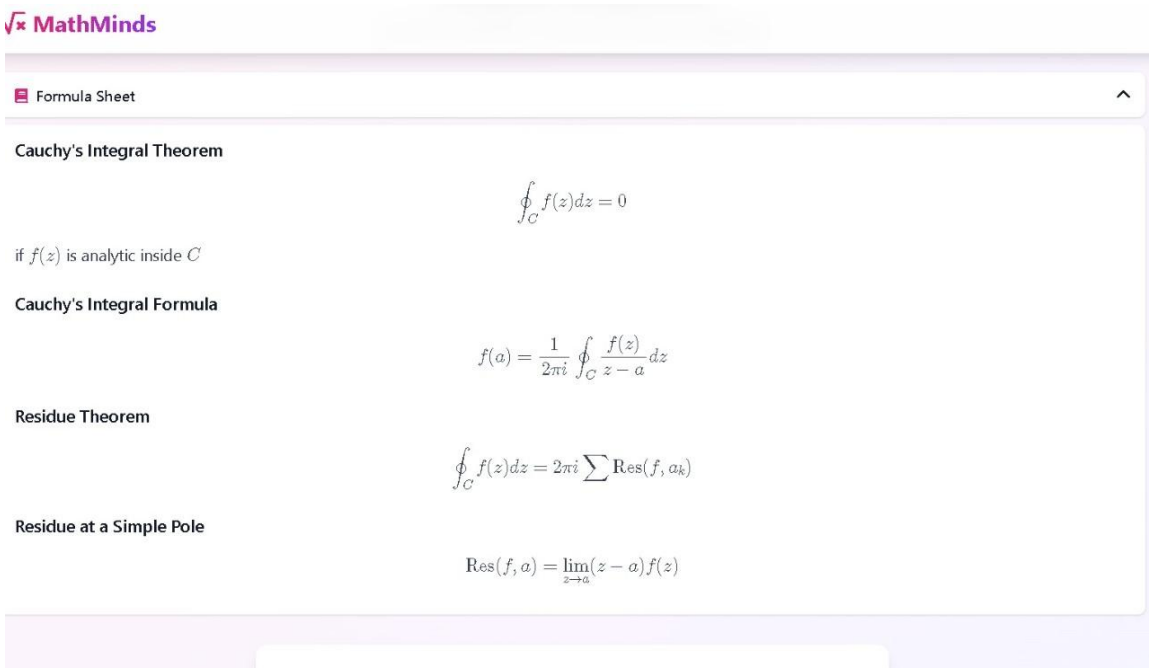

Figure 8: Formula Sheet

The formula sheet in the system acts as a quick reference module that provides users with important mathematical formulas and theorems in a structured format. It helps students easily access concepts such as Cauchy's Integral Theorem, Cauchy's Integral Formula, and Residue

Theorem without searching externally. The formulas are displayed in properly formatted mathematical notation (LaTeX), improving readability and understanding. This feature supports learning by reinforcing theoretical knowledge alongside problem-solving. Additionally, it assists users in verifying steps and selecting appropriate formulas while solving complex problems.

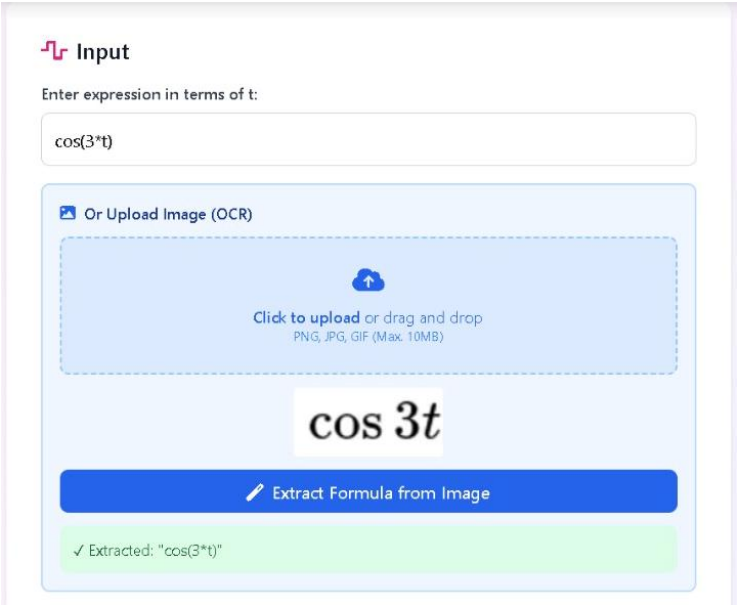

Figure 9: Image Input

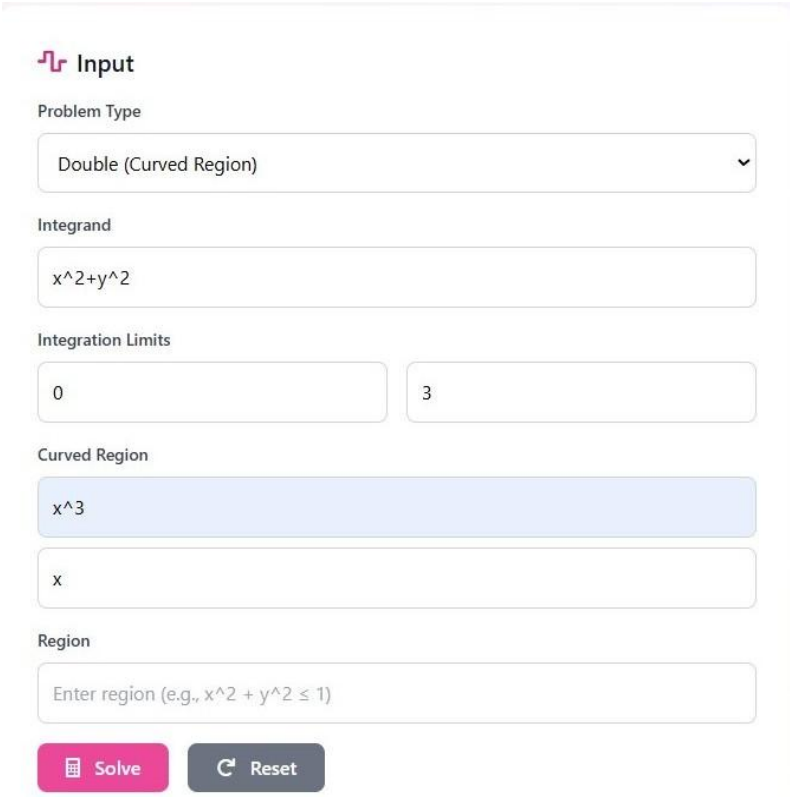

Figure 10: Text input for Double Integration Problem

Here we have chosen the module double integration, and are solving double integrals over a curved region.

The following equation is first transformed into its corresponding LaTeX format.

$$\int_0^3 \int_{x^3}^{x} (x^2 + y^2)\, dy\, dx$$

It is then represented in LaTeX format as:

$$\{\backslash int_\{0\}\hat{}\{3\}\backslash int_\{x\hat{}\{3\}\}\hat{}\{x\}(x\hat{}\{2\} +\backslash y\hat{}\{2\})\backslash, dy\backslash, dx\}$$

The LaTeX format of the equation is then forwarded to the backend logic generator for further processing.

Solution

$\int_{x^3}^{x} x^2 + y^2\, dy = -\frac{x^9}{3} - x^5 + \frac{4x^3}{3}$

$\int_0^3 -\frac{x^9}{3} - x^5 + \frac{4x^3}{3}\, dx = -\frac{10314}{5}$

Final Answer: $-\frac{10314}{5}$

Figure 11: Generated Step-wise Solution

In the first step the backend system performs integration with respect to 'y' which is the inner integral

$$\int_{x^3}^{x} (x^2 + y^2)\, dy = \left[x^2 y + \frac{y^3}{3}\right]_{x}^{x^3}$$

After applying the limits y = x3 to y = x:

$$= \left(x^3 + \frac{x^3}{3}\right) - \left(x^5 + \frac{x^9}{3}\right) = \frac{4x^3}{3} - x^5 - \frac{x^3}{9}$$

In the second step the system performs integration with respect to 'x' which is the outer integral

$$\int_0^3 \left(\frac{4x^3}{3} - x^5 - \frac{x^3}{9}\right) dx = \left[\frac{x^4}{3} - \frac{x^6}{6} - \frac{x^{10}}{30}\right]_0^3$$

Apply limits to the equation as x = 3 to x = 0:

$$= 27 - \frac{729}{6} - \frac{59049}{30}$$

Take the common denominator as 30:

$$= \frac{810 - 3645 - 59049}{30} = \frac{-61884}{30} = \frac{-10314}{5}$$

The final result of the problem is then showcased onto the interface.

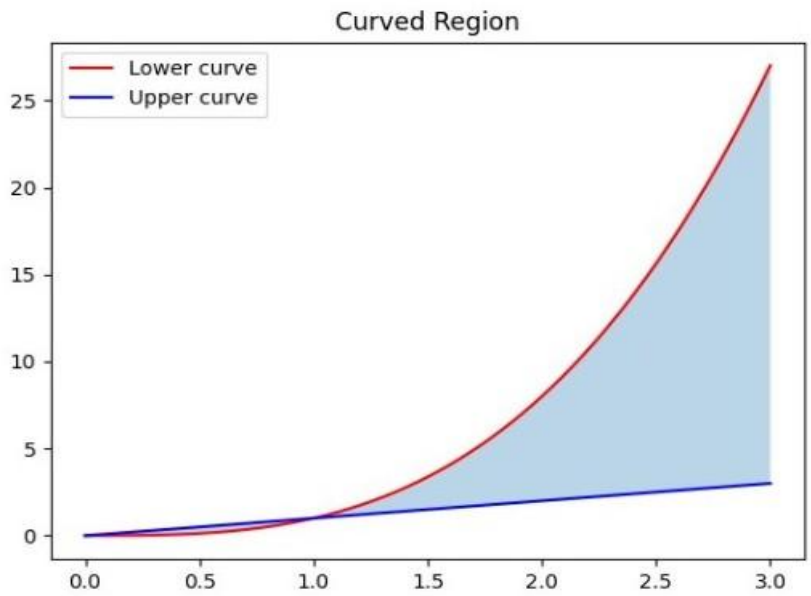

Figure 12: Area Under Curved Region

The figure presents the evaluation of a double integral over a curved region defined by the functions y=x3 and y = x within the interval 0≤x≤3. The shaded plot illustrates the region of integration, and the corresponding analytical solution is obtained by sequential integration with respect to y and x. The resulting negative value reflects the reversal of upper and lower bounds over the specified domain.

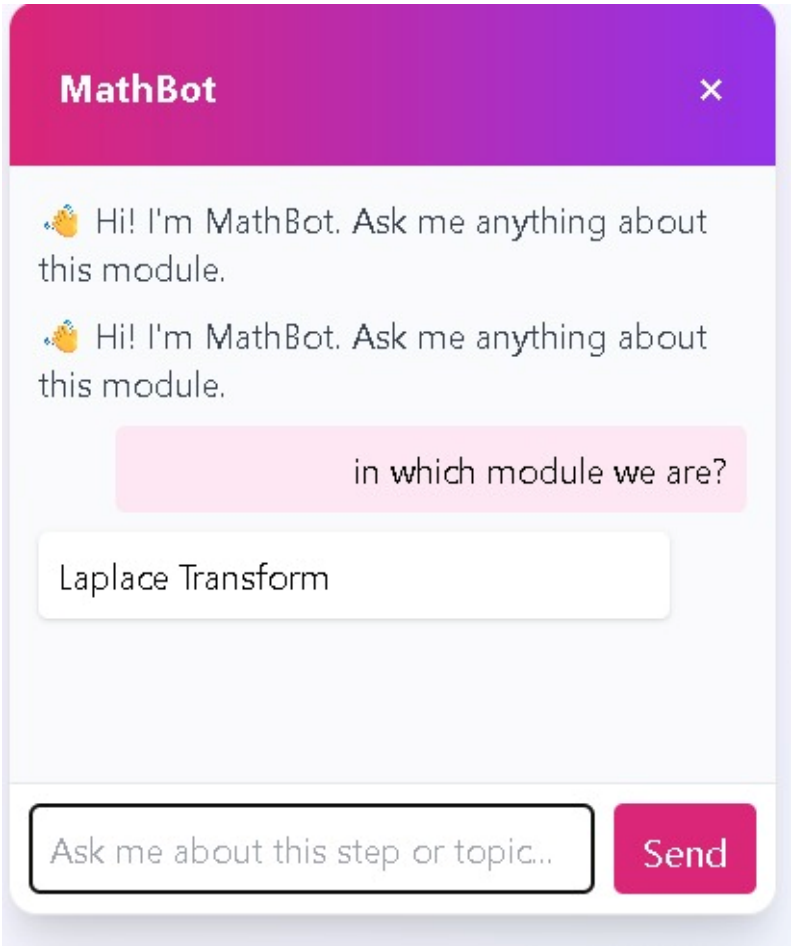

Figure 13: MathBot Interface

Finally, the chatbot interface (MathBot) is a user-friendly platform where users can ask queries related to specific mathematical modules or solution steps. It supports interactive communication by allowing users to type questions and receive instant, context-aware responses. This implementation enhances user engagement and helps in better understanding of concepts through real-time assistance.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The project effectively addresses one of the major challenges faced by engineering students understanding and solving complex problems in Engineering Mathematics (EM1–EM4). Many students struggle to grasp fundamental concepts and step-by-step solution methods, often relying solely on final answers without clarity on the reasoning behind them. This platform bridges that gap by providing detailed, concept-oriented explanations that guide learners through each stage of problem solving. By offering topic-wise modules and personalized doubt-solving support, it transforms the traditional learning experience into an interactive and self-paced process. The system helps students strengthen their conceptual understanding, build confidence in applying mathematical methods, and reduce academic dependency on external coaching or textbooks. Ultimately, this project serves as a digital learning companion designed to enhance analytical thinking and improve academic performance in core Engineering Mathematics subjects.

The existing platform has certain limitations. The solution mechanism implemented in the

system is rule-based and relies on predefined solution paths which restricts the system's ability to handle unseen or highly complex problems. While this approach ensures correctness, the system gets limited as it is not scalable across diverse mathematical domains, and it only works on the modules relevant to the curriculum. Additionally, the correctness of the solution is influenced by the quality of input questions.

8.2 Future Work

This project offers significant potential for future enhancements. The system can be integrated with intelligent solution engines which are capable of solving unseen Mathematical problems using machine learning models. Adaptive learning models can be introduced to personalize the concepts based on the user's strengths and weaknesses and based on these each user will have their own personalized learning experience. Expanding the sets of problems which can be solved by the user can be extended and topics like advanced numerical methods, higher-order differential equations etc. can be added to attract a more varied userbase. With these enhancements, the system can evolve into a comprehensive intelligent tutoring solution for engineering education.

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Publication

The paper entitled "**MathMinds: A StepWise Solution Generator For Engineering Mathematics Using OCR**" has been presented at "**2nd International Conference on Emerging Technologies and Computing Innovations (ICETCI-2026)**" by "Prof. Vijaya Bharathi Jagan, Kartik Parmar, Dhruv Wesavkar, Niraj Patel, Rohan Patil ". The paper was reviewed and accepted by the ICETCI 2026 Organizing Committee and is scheduled for inclusion in the conference proceedings published by **Springer Nature**.

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for the paper titled

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MathMinds: A Stepwise Solution Generator for Engineering Mathematics Problems Using OCR

In appreciation of his/her exceptional achievement and contribution in the 2nd International Conference on Emerging Technologies and Computing Innovations (ICETCI 2026), organized by MG Aricent Educational Foundation, Nashik, India on March 21 & 22 , 2026 (Hybrid Mode)

Dr. Pradeep M Jawandhiya
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Math Minds: A Stepwise Solution Generator for Engineering Mathematics Problems Using OCR

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Abstract. Engineering Mathematics (EM1–EM4) are core subjects in the Mumbai University curriculum, yet many students struggle with its complex problem-solving methods. This project acts as a platform for studying mathematics, resolve doubts and making learning easier. Students can either type the required problem or upload questions via screenshots or photos, which is then processed using Optical Character Recognition (OCR) to extract mathematical expressions. The extracted equations are converted into LaTeX format and processed using a symbolic mathematics engine to generate step-by-step solutions. Unlike traditional problem-solving tools that provide only final answers, the proposed system focuses on generating detailed intermediate steps to help students understand the solution process. The system then generates step-by-step solutions using rule-based algorithms and symbolic mathematics engine, helping students grasp concepts rather than just answers. The platform also offers topic-specific practice for all the modules. The project intends to function as a customized digital instructor by fusing AI with an interactive interface, lowering academic challenges and enhancing EM1–EM4 students' problem-solving abilities

Keywords: Optical Character Recognition (OCR),Stepwise solutions ,Digital tutor, LaTeX.

1 Introduction

Rapid scientific and engineering advancements in the modern world necessitate accurate and efficient solutions to challenging mathematical problems. Numerous technical disciplines, including computer science, electronics, mechanical design, and system engineering, are built on mathematics. However, lengthy analytical derivations, symbolic manipulations, and complex mathematical problems involving Fourier series, Laplace transforms, differential equations, and matrix computations can be challenging for both professionals and students. Manual fixes for these issues take a lot of time and could result in human error.

By offering an intelligent platform that can comprehend and solve mathematical expressions step-by-step, the suggested system "MathMinds: A Stepwise Solution Generator for Engineering Mathematics Problems using OCR" seeks to close the gap between conventional problem solving and computational automation [2][5]. Both text-based and image-based inputs are supported by the system. Optical Character Recognition (OCR) methods are used to process image inputs, extracting mathematical expressions and transforming them into structured formats appropriate

for symbolic computation.

The implementation creates an end-to-end processing pipeline by integrating multiple Python-based technologies. Mathematical expressions can be identified from both printed and handwritten images using PaddleOCR and EasyOCR [12]. Pix2Text allows for the structured representation of mathematical expressions by converting the extracted equations into LaTeX format. The SymPy library carries out symbolic operations like algebraic simplification, differentiation, integration, and transform evaluation after the equation has been transformed into a machine-readable format [6]. While MathJax is used to render the final solutions in an understandable mathematical format, supporting libraries like Pillow and NumPy help with image preprocessing and numerical operations. Through the integration of computer vision, artificial intelligence, and symbolic mathematics, the platform serves as both an automated solver and a learning tool that provides step-by-step solutions. Instead of depending only on definitive solutions, this method fosters conceptual understanding.

All things considered, the suggested platform shows how contemporary computational methods can make solving challenging mathematical problems easier while preserving interpretability and transparency. The system offers a scalable framework for intelligent mathematics learning and automated problem solving in engineering and research settings by fusing OCR technologies with symbolic computation.

2 Related Work

The main motivation behind this project emerged after research on the automated mathematical problem-solving systems and learning platforms. We studied previous projects such as SOLVE IT(MATH SOLVER) and Wolfram alpha these works showcased us how automation reduces manual problem-solving efforts while also simultaneously teaching its users. These systems, showed strong symbolic computation capabilities and relying in solving complex mathematical problems, but they either lacked support for image-based inputs or have low accuracy for image-based inputs[2].This limited their usability for students who relied on their handwritten notes, question papers or textbook images. These highlighted a gap to us in the existing systems, which motivated us to create a more robust solutions which would solve these problems.

Advancements in educational technology emphasized the role of Intelligent Tutoring Systems (ITS) in improving learning outcomes. Systematic reviews and meta-analysis reported that the ITS platforms enhance students' performance through adaptive instructions and self-paced learning environments [5][4]. Research on web-based mathematics instruction further reinforced these findings, demonstrating that interactive and technology-driven learning environments enhance problem-solving skills, engagement, and accessibility across diverse learner groups [1]. These studies collectively emphasized that effective educational systems must move beyond static content delivery toward learner-centric and adaptive designs.

Previous research faced challenges for accurate recognition of mathematical problems. Traditional OCR techniques fail to preserve the two-dimensional structure of mathematical notations. To address this issue, recent studies proposed expression trees and convolutional sequence networks which are a type of structured representations of mathematical problems, and this helps maintain the mathematical validity and operators' hierarchy during equation generation [12][11]. These helped improve reliability and interpretability of machine generated equations.

Symbolic computation tools enabled the students to connect theoretical derivations with computational verification, which further enhanced the analytical and reduced conceptual gaps [7][3]. These findings reinforced importance of step-by-step derivations rather than black-box numerical outputs.

Recent studies on multimodal mathematical reasoning highlighted the potential of combining visual, textual and symbolic inputs for solving various mathematical problems [10]. At the same time, the research on handwritten mathematical expressions showed progress while using encoder-decoder architecture, graph neural networks and self-supervised attention mechanisms, while acknowledging persistent challenges in generalization and interpretability [9][8].

3 Methodology

The proposed system follows a systematic approach to aid students in resolving Engineering Mathematics problems (EM1–EM4) while promoting conceptual understanding. At first, the user chooses the type of math module and the problem can be given as text or as an image by taking a screenshot.
To make the input images easier to read, they are preprocessed. This makes it possible to use OCR techniques to accurately extract mathematical expressions[4]. The extracted content is then changed into the right LaTeX format so that it can be structured and processed further.

Then, the system uses standard university-level methods to match the problem to a set of pre-defined solution strategies. Solutions are produced sequentially, akin to classroom derivations, guaranteeing clarity and accuracy [6].

Finally, the results are formatted into readable mathematical expressions and displayed through a web interface [12]. The platform also organizes problems into topic-based modules to support guided practice and self-paced learning, helping students better understand complex mathematical concepts.

3.1 Proposed System

The proposed system follows a structured architecture to ensure modularity, clarity, and efficient processing of Engineering Mathematics problems. The system handles both textual and image based inputs and then provide structured step-by-step solutions.
User Interface Layer. The user interface layer acts as the interaction module between the user and the system. It allows the users to either manually enter mathematical expressions or they can upload images containing handwritten or printed equations as shown in Fig 1.

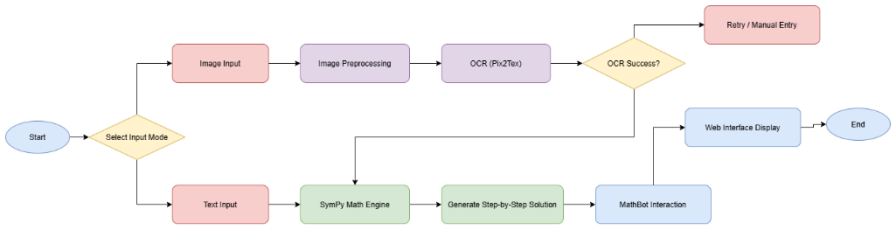

Fig. 1. Proposed System.

Fig. 2. LaTeX Conversion of Mathematical Equation

The system supports two modes of user input: text-based and image-based. When a user enters a mathematical expression in text form, it is first converted into a standardized LaTeX representation to ensure uniformity and accurate symbolic interpretation which is shown in Fig.2. In case of image-based input, the image is first processed to extract the mathematical expression, which is later converted to its corresponding LaTeX form[11]. The system converts all equations to its LaTeX form to establish a consistent intermediate representation for further processing.

Backend Layer. It is the central control unit of the proposed system. It receives input from either the user interface or the OCR module through a request handler, which is responsible for managing request routing, validation and error handling. This ensures that only well-formed and interpretable inputs proceed to subsequent stages to be processed.

An equation parser analyzes the extracted text to determine the specific mathematical operation involved, such as integration, Laplace transform, or matrix computation. Based on the analysis, the extracted expressions are arranged into a standardized format, which is suitable for further computation.

Math Engine Layer. It performs the core computational tasks. Using symbolic computation techniques, the engine executes the required mathematical operations. Additionally, a step-by-step logic generator decomposes the solution into intermediate stages, enabling users to understand the complete problem-solving methodology rather than just the final result [7].

Finally, the Output Module formats the computed result and its corresponding step-by-step explanation into a readable mathematical form. The formatted output is sent back to the user interface, where it is displayed using mathematical rendering tools to ensure clarity, correctness, and ease of interpretation from Fig.3.

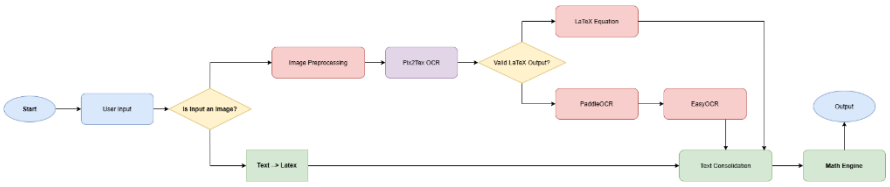

Fig. 3. Working of OCR Models

OCR Models.

PaddleOCR. It is an OCR framework for general text detection and recognition that is based on deep learning. It works well with structured mathematical expressions and legible printed text that are frequently found in scanned documents, PDFs, and textbooks. Text regions are first identified and then transformed into machine-readable strings using PaddleOCR detection–recognition pipeline. It might, however, have trouble with intricate spatial structures found in sophisticated mathematical notation.

Pix2Tex. It is a specialized model created especially to convert mathematical expression images into LaTeX format. To capture the spatial arrangement of symbols, it employs an encoder-decoder architecture with a vision transformer-based encoder and a sequence generation decoder. Because of this, it is better at identifying complicated equations that contain fractions, matrices, superscripts, and subscripts.

EasyOCR. It is used for more difficult inputs, which include examples of handwritten equations or images taken under low-light conditions. It helps improve the overall robustness of the system by being able to interpret text that is not well-formed (unusual) or very noisy from an image taken with a smart device. In addition, while its ability to comprehend complex mathematical structures is limited, it provides a fallback for cases when other OCR methods are not able to yield results.

Chatbot. The chatbot operates through a layered workflow comprising the frontend, backend task logic, and the Gemini API, as illustrated in the Fig.4. The user first submits a query or selects the predefined steps through the chatbot interface. The input is captured and forwarded to the backend for processing. In the backend task layer, the system extracts the relevant question, and then constructs a query by combining the user's query and the system information. This prompt is then sent to the Gemini API for further processing.

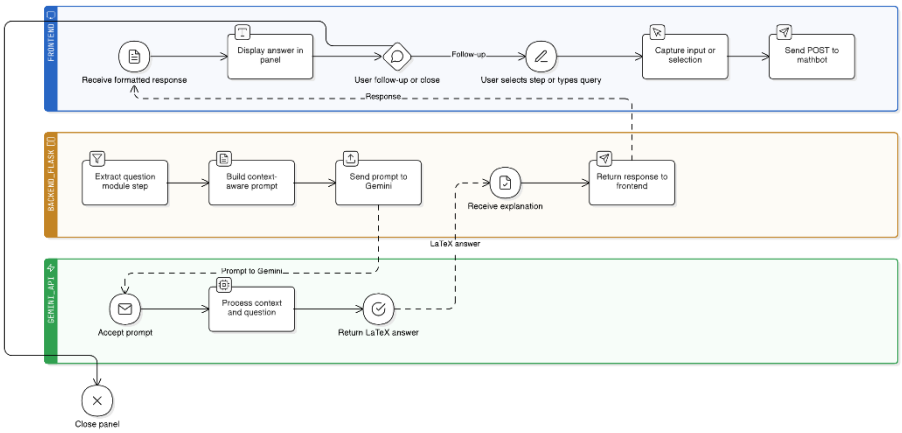

Fig. 4. Workflow Of Gemini API.

Upon receiving the generated prompt the Gemini API analyses it and then generates a LaTeX formatted explanation. This generated response is then forwarded to the backend, which validates and formats it properly before forwarding it to the frontend. The system also supports follow-up interactions, so the user can enquire further about the steps. This workflow continues iteratively until the user themselves choose to close the panel this ensures that there is a seamless interaction between the user and the system.

3.2 System Overview

This system is organized around the Mumbai University (MU) curriculum for the subjects Engineering Mathematics I, II, III, IV. Each of these subjects are having six different modules being taught in the undergraduate engineering program. These modules each cover a vast range of concepts spanning from differential equations, matrices, probability, Laplace transform etc. This helps the users to navigate concepts systematically. Also, in each module there is a dedicated formula sheet which is provided which provides the users a quick access to important formulas and key concepts for their revision and reference. This feature helps the users to quickly recall the fundamental concepts while solving problems, which further improves their problem solving capabilities.

3.3 Handling OCR Errors

The correctness of the mathematical expression extracted during the OCR stage determines the accuracy of symbolic computation in the suggested system. In real-world situations, handwriting variations, image noise, or intricate mathematical layouts can occasionally cause OCR models to misinterpret symbols. The system incorporates a user validation step prior to the computation stage in order to address this problem. The interpreted equation appears on the user interface once the OCR module has extracted the equation and transformed it into the appropriate LaTeX representation. If any recognition errors are found at this point, the user has the option to manually edit to fix the equation. The equation is sent to the symbolic computation engine for step-by-step solution generation only upon user confirmation. This human-in-the-loop verification method guarantees greater reliability of the produced mathematical derivations and greatly lessens the impact of OCR recognition errors on the final solution. The system maintains accuracy and flexibility for various input formats, including handwritten notes, scanned question papers, and screenshots, by permitting user correction.

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4 Dataset Used

We have utilized the IM2LATEX-100K dataset, which is a large-scale, prebuilt benchmark dataset designed for image-to-LaTeX generation tasks. It contains approximately 100,000 paired samples, where each sample consists of a rendered mathematical formula image and its corresponding LaTeX source code. Unlike the

traditional datasets, IM2LATEX-100k does not contain predefined numerical features, instead the model learns these features automatically by receiving raw inputs:
Image Features: Each sample contains rendered formula images. Pixel level data is extracted using Convolutional Neural Networks (CNNs) while model training, which helps the model capture the structural information of the Mathematical expressions such as the spatial alignment, symbols and Mathematical layout.
Textual Features: The output is recorded as a sequence of LaTeX tokens, these tokens are then converted into their corresponding embeddings, where each token represents a learned feature from the high dimensional space.

Feature Dimensionality: These are dependent on the model used and vary based on the architecture used in the CNN encoder and sequence decoder. Hence, the dataset itself does not have a fixed feature count.

The key dataset Characteristics and Statistics used in this study is summarized in Table 1.

Table 5. Dataset Characteristics and Statistics.

Attributes	Details
Total Samples	103,558 image-formula pairs
Training Samples	90,000
Validation Samples	5000
Test Samples	5000
Image Resolution	160x320 pixels
Average Formula Length	30-40 LaTeX Tokens

5 Model Evolution

Table 6. Performance Comparison of OCR Models.

Metrics	PaddleOCR	EasyOCR	Pix2Tex
Character Recognition Accuracy (%)	92.4	78.1	96.8

Precision (%)	78.6	72.3	95.9
Recall (%)	91.2	76.4	95.1
F1-Score (%)	90.1	78.8	95.5
LaTeX Parsing Success Rate (%)	81.3	74.5	92.7

As shown in Table 2 , Pix2Tex performs better than the other general OCR models in every expression-level statistic, making it ideal for Engineering Mathematics tasks involving complex notations.

Although PaddleOCR has high character-level accuracy, it has trouble maintaining mathematical structure, especially when dealing with multi-line or nested formulas.

EasyOCR works well for simple text recognition, but it struggles to parse complex mathematical symbols and portray them accurately. Higher LaTeX parsing success rates directly correlate with improved symbolic computation reliability and step-by-step solution generation.

5.1 Experimental Setup

The suggested system uses both pretrained models and experimentally evaluated baseline models for OCR and image-to-LaTeX. In order to assess the efficacy of deep learning architectures for mathematical expression recognition, samples from the IM2LATEX-100K dataset were used to implement and train CNN+RNN and CNN+LSTM. These experiments yielded the performance metrics reported for these baseline models.

However, the obtained accuracy was lower compared to specialized mathematical OCR systems. Therefore, pretrained models such as PaddleOCR, EasyOCR, and Pix2Tex were integrated into the final system pipeline. The benchmark results recorded in the corresponding research publications and official implementations serve as the foundation for the performance metrics presented for these models. In order to give a comparative overview of the efficacy of various OCR techniques for tasks involving the recognition of mathematical expressions, these metrics are included.

6 Results

The main goal of the proposed system is to help students learn Engineering Mathematics by correctly interpreting the equations they send in and giving them step-

by-step symbolic solutions. It also needs to keep the math structure, work with different types of input, and work with symbolic computation. Fractions, matrices, integrals, and summations are common two-dimensional layouts in engineering math problems. Pix2Tex changes pictures of equations into structured mathematical forms while keeping the spatial hierarchy and operator precedence. This makes it better than regular OCR engines, which treat equations as linear text and often lose structural information.

Table 7. Table captions should be placed above the tables.

Model	Accuracy	Precision	Recall	F-1 score
TesseractOCR	67	65	58	61
InftyReader	87	88	82	85
Im2LaTeX	92	91	89	90
Mathpix	94	94	93	93
CNN + RNN	81	70	77	78
CNN + LSTM	80	78	79	78
PaddleOCR	88	88	84	86
EasyOCR	79	80	76	78
Pix2Tex	92	93	91	92

As shown in Table 3 there are other high-accuracy systems like MathPix, but they need cloud access or proprietary APIs, which makes it hard to reproduce and use them for academic work offline. PaddleOCR and EasyOCR, on the other hand, are good opensource options that work well on printed and semi-structured equations.

They also make the system faster and more scalable. Students send in equations in a number of different ways, such as typed text, screenshots, and handwritten notes. No single OCR model works best in all situations, so combining general-purpose OCR engines with a math-aware recognition model makes it more robust against changes in handwriting, image quality, and formatting.

The symbolic computation engine (SymPy) used in the system works with the outputs of these models to make it possible to find solutions step by step. This method keeps

things clear and helps the platform reach its educational goal of being a learning tool instead of a black-box solver.

6.1 Experimental Results

After opening the MathMinds interface, as student can navigate through semester-wise engineering Mathematics module and can select their required subject.
After selecting a subject, the student is presented with all the major engineering Mathematics topics. From here on the student can choose a specific module such as Eigen vector and values, complex integration, Z-Transform, Inverse Z-transform, Linear programming etc.

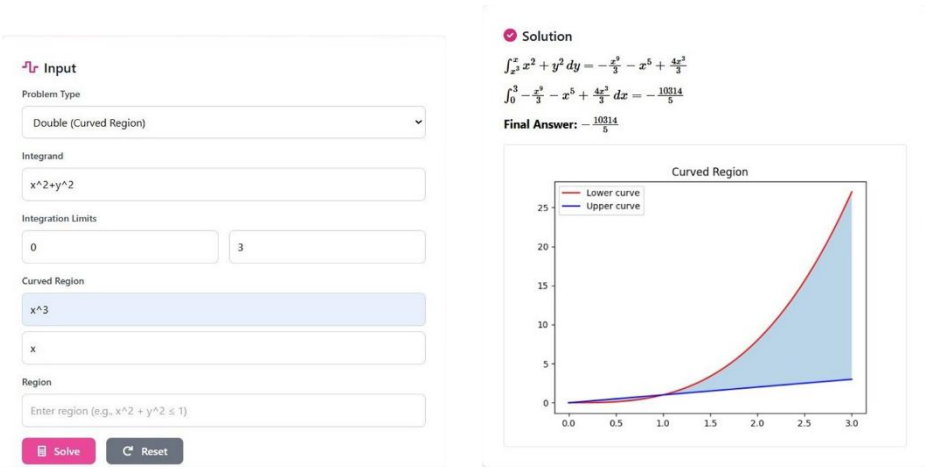

Fig. 5. Input and final solution for the double integration problem.

The Fig.5. illustrate the problem solving steps for double integration double integrals over a curved region.

$$\int_0^3 \int_{x^3}^{x} (x^2 + y^2)\, dy\, dx \tag{1}$$

The above equation gets converted into its corresponding LaTeX format. The LaTeX format of the equation is then forwarded to the backend logic generator. In the first step the backend system performs integration with respect to ‘y’ which is the inner integral.

$$\int_{x^3}^{x} (x^2 + y^2)\, dy = \left[x^2 y + \frac{y^3}{3} \right]_{x^3}^{x} \tag{2}$$

After applying the limits y = x^3 to y = x:

$$= \left(x^3 + \frac{x^3}{3} \right) - \left(x^5 + \frac{x^9}{3} \right) = \frac{4x^3}{3} - x^5 - \frac{x^9}{3} \tag{3}$$

In the second step the system performs integration with respect to 'x' which is the outer integral.

$$\int_0^3 \left(\frac{4x^3}{3} - x^5 - \frac{x^9}{3} \right) dx = \left[\frac{x^4}{3} - \frac{x^6}{6} - \frac{x^{10}}{30} \right]_0^3 \tag{4}$$

Apply limits to the equation as x = 3 to x = 0:

$$= 27 - \frac{729}{6} - \frac{59049}{30} = -\frac{10314}{5} \tag{5}$$

The final result of the problem is then showcased onto the interface.

The final output presents the evaluation of a double integral over a curved region defined by the functions y = x3 and y = x within the interval $0 \leq x \leq 3$. The shaded plot illustrates the region of integration, and the corresponding analytical solution is obtained by sequential integration with respect to y and x. The resulting negative value reflects the reversal of upper and lower bounds over the specified domain.

7 Conclusion

Using OCR and symbolic computation, the suggested system offers an intelligent learning platform for resolving Engineering Mathematics (EM1–EM4) problems. Instead of depending solely on final answers, the platform allows students to obtain structured step-by-step solutions that improve conceptual understanding by supporting both text-based and image-based inputs. Problem interpretation and solution generation are enhanced when OCR models are integrated with symbolic computation. All things considered, the system shows how AI-driven tools can enhance mathematics education by making difficult problem solving simpler while preserving interpretability and transparency.

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